



Serkan Sulun, PhD

Machine learning scientist

sulun.org

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PUBLICATIONS

Sulun, S. (2026). Dance-to-Music Generation: A Short Survey and Open Problems

Sulun, S., Viana, P., & Davies, M. E. P. (2026). Video Soundtrack Generation by Aligning Emotions and Temporal Boundaries. *IEEE Transactions on Multimedia*. (*in print*) [arXiv](#) | [code](#)

Sulun, S. (2025). Video-based Music Generation, PhD Thesis. [arXiv](#) | [code](#)

Sulun, S., Viana, P., & Davies, M. E. P. (2024). VEMOCLAP - A video emotion classification web application. 2024 IEEE International Symposium on Multimedia (ISM), 137–140. [DOI](#) | [arXiv](#) | [demo](#)

Sulun, S., Viana, P., & Davies, M. E. P. (2024). Movie trailer genre classification using multimodal pretrained features. *Expert Systems with Applications*, 258, 125209. [DOI](#) | [arXiv](#) | [code](#)

Sulun, S., Oliveira, P., & Viana, P. (2023). Emotion4MIDI: A Lyrics-based Emotion-Labeled Symbolic Music Dataset. *EPIA 2023. Lecture Notes in Computer Science*, vol 14116. Springer. [DOI](#) | [arXiv](#) | [code](#)

Sulun, S., Davies, M. E. P., & Viana, P. (2022). Symbolic Music Generation Conditioned on Continuous-Valued Emotions. *IEEE Access*, 10, 44617–44626. [DOI](#) | [arXiv](#) | [code](#)

Sulun, S., & Tekalp, A. M. (2021). Can learned frame prediction compete with block motion compensation for video coding? *Signal, Image and Video Processing*, 15(2), 401–410. [DOI](#) | [arXiv](#) | [code](#)

Sulun, S., & Davies, M. E. P. (2020). On filter generalization for music bandwidth extension using deep neural networks. *IEEE Journal of Selected Topics in Signal Processing*, 15(1), 132–142. [DOI](#) | [arXiv](#) | [code](#)

Sulun, S. (2018). Deep Learned Frame Prediction for Video Compression, Master's Thesis. [arXiv](#) | [code](#)

WORK EXPERIENCE

Computer Vision Center, Postdoctoral researcher

May 2026 – Present

Barcelona, Spain

INESC TEC, Researcher

Mar. 2019 – Mar. 2026

Porto, Portugal

- Automatic video-based symbolic music generation
- Audio bandwidth enhancement
- Finetuning vision language models for industrial safety

University of Porto, Lecturer

Sep. 2023 – Present

Porto, Portugal

- Lecturer for the course Programming with Python

Catolica Porto Business School, Lecturer

Mar. 2024 – Present

Porto, Portugal

- Lecturer for the course Python for Business Analytics

Koc University, Research assistant

Sep. 2016 – Nov. 2018

Istanbul, Turkey

- Video compression and frame prediction using deep neural networks

Koc University, Teaching assistant

Sep. 2016 – Nov. 2018

Istanbul, Turkey

- Preparing and leading laboratory sessions for the course, Programming with MATLAB

Champalimaud Centre for the Unknown, Intern

Jul. 2013 – Oct. 2013

Lisbon, Portugal

- Processing neural signals and videos obtained by rat experiments
- Designing the controller board. ▪ Article acknowledgment: [DOI](#)

Sabancı University

Feb. 2010 – Jun. 2014

Istanbul, Turkey

- Peer lecturer on freshman mathematics, physics, chemistry, and biology

MANAGEMENT AND LEADERSHIP

Principal Investigator: INESC TEC Seed project: Video and MIDI processing using deep neural networks.

Supervisor: Master's thesis: Emotion classification of MIDI lyrics using transformers

Supervisor: Bachelor's project: Speech emotion classification

Coordinator: Lab sessions and 5 assistants in the course, Introduction to MATLAB

EDUCATION

University of Porto

PhD, Electrical and Computer Engineering

Present

Porto, Portugal

Koc University

MS, Electrical and Electronics Engineering (GPA: 3.81/4, converted to 19/20)

Nov. 2018

Istanbul, Turkey

Sabanci University

BS, Electronics Engineering (GPA: 3.86/4, converted to 20/20)

Jun. 2014

Istanbul, Turkey

SKILLS

Deep learning: Pytorch, Tensorflow, generative AI, transformers, adversarial learning, convolutional neural networks, vision language models, LoRA finetuning, cloud platforms (AWS, GCP, Azure), Slurm

Programming: Python, MATLAB, C++, Julia

Signal processing: MIDI, audio, image, video, natural language processing; video compression; computer vision

Miscellaneous: Machine learning, remote development, Git, LaTeX, Ubuntu, Bash

Music: Drumming, Cubase, Protools, Audacity, MIDI

INVITED TALKS

Introduction to deep learning, *Polytechnic Institute of Porto, Portugal*

Deep learning for music generation, *University of Porto, Portugal*

Deep learning for music information retrieval, *University of Porto, Portugal*

INTERVIEWS

SIC Noticias: Portugal vai receber Turquia

[Website](#) | [YouTube](#)

Observador: Generating music from videos

[Portuguese original](#) | [English translation](#)

Olhar No Progresso: Deep learning, multimedia, and music

[Website](#) | [YouTube](#)

LANGUAGES

English (C2), Portuguese (B1), Spanish (A1), Turkish (native)

ACTIVITIES

Finished in 04:20:05, in the 42 km. Porto Marathon

Drummer, in Santa Cecilia Music School rock band

Parachuting, in Turkish Aeronautical Association with 5-day training camp and 400 m. solo jump

Volunteer, Telfs Youth Center Austria (Erasmus+) and World Wide Opportunities on Organic Farms (WWOOF)